# Understanding Transferable Urban Mobility Structure: Distance Decay Recovery and Interpretable Zero-Shot Prediction

Thinh Nguyen[1*]

[1*]Aarista Technologies, Ho Chi Minh City, Vietnam.

Corresponding author(s). E-mail(s): nqt900@gmail.com;

**Abstract**

Developing transferable mobility models for cities without origin-destination (OD) observations remains an open challenge. Existing approaches typically rely on historical OD records, travel surveys, or individual mobility traces. However, these data sources are often expensive to collect, difficult to obtain, and subject to administrative, privacy, and regulatory constraints. To address these challenges, we introduce the Separable Decomposed Framework for Zero-Shot Transferable Flows (SDF-ZTF), a decomposed mobility modeling approach that factorizes OD flows into three interpretable components: target-recovered distance decay, open-data-driven destination attraction, and cross-city transferable origin production. By relying exclusively on globally available open datasets, including OpenStreetMap and WorldPop gridded population data, this methodology enables mobility estimation for previously unseen cities without requiring local OD observations, surveys, or target-city retraining. A key contribution of this work is a cross-city feature transferability analysis that identifies a compact set of stable urban indicators capable of generalizing across heterogeneous cities. The decomposed architecture not only improves interpretability by isolating the contribution of each mobility mechanism but also provides a practical pathway for diagnosing model limitations and guiding future improvements. Experimental results demonstrate that our approach achieves performance comparable to state-of-the-art deep learning models under fully zero-shot transfer settings while maintaining transparency and policy relevance.

**Keywords:** Urban mobility, origin-destination flows, gravity models, interpretability, transfer learning, zero-shot transfer, open-data modeling

## 1 Introduction

Urban mobility modeling is a critical pillar for modern infrastructure planning, epidemiological tracking, and transportation management (Barbosa et al. 2018; González et al. 2008; Song et al. 2010). Accurately predicting origin-destination (OD) flows within a city is essential for addressing municipal challenges ranging from congestion mitigation to transit-oriented development. However, traditional approaches for collecting OD matrices rely on expensive, large-scale household travel surveys that suffer from decade-long update cycles, resulting in persistent data gaps, particularly in developing regions.

To address these data gaps, human mobility researchers have proposed two main paradigms. Classical spatial interaction models, such as gravity and radiation models, assume a structured

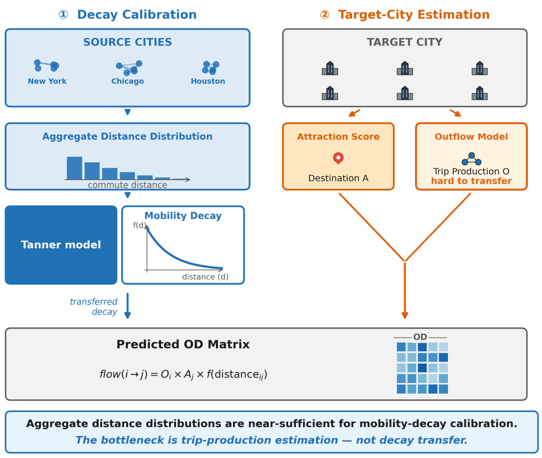


**Fig. 1** SDF-ZTF framework: Three independent components (origin production $O_i$ from features, distance decay $f(d)$ from aggregate data, destination attraction $A_j$ from features)

relationship where travel flows are determined by physical proximity and zone attributes (Wilson 1971; Simini et al. 2012; Stouffer 1940). Modern deep neural models, such as DeepGravity and graph neural networks, treat mobility prediction as a monolithic learning task, mapping high-dimensional urban features to flow rates using deep multi-layer architectures (Simini et al. 2021; Rong et al. 2023; Wang et al. 2021).

Despite their success, existing paradigms face a fundamental trade-off: **interpretability versus transferability**. Classical gravity models provide clear, policy-relevant interpretations but perform poorly when transferred to new cities without local calibration. Conversely, deep neural models achieve high predictive accuracy but function as black boxes, sacrificing structural transparency and requiring extensive local, target-city OD data for training or fine-tuning. Whether an interpretable, structured model can generalize to unseen cities without local mobility observations remains an open question.

This represents a critical research gap in spatial interaction modeling. In the modern era, the ubiquity of mobile devices and social networks enables technology corporations to construct highly accurate geographic movement probability distributions of individuals based on opt-in location-sharing data. While certain demographics like the elderly and children may not interact directly with these digital platforms, they represent a minor share of overall daily mobility; the vast majority of active commuters—namely working professionals and students—are heavily active on social media and mobile communication channels for work and leisure. This digital footprint offers a novel source for mobility modeling. However, utilizing such raw individual-level trajectories directly raises significant privacy and administrative concerns.

Thus, a key question is whether we can leverage these aggregate mobility probability distributions (or trip-length distributions) provided by social networks and mobile providers to recover and calibrate the distance decay function without exposing sensitive individual trajectories. Furthermore, we must understand the relative importance of the components: destination attraction, distance decay, and origin production, and identify which of these elements exerts the dominant influence on predictive accuracy.

To address these challenges, we investigate three core research questions:

1. Can travel probability distributions derived from location-sharing mobile devices and social networks—which capture the active commuting population—be utilized to effectively estimate and calibrate the distance decay function?
2. Of the three fundamental components of spatial interaction (destination attraction, distance decay, and origin production), which is the most critical factor for accurate flow estimation?
3. Can a decomposed model utilizing transferable built-environment features identified from globally available open data achieve zero-shot transfer performance comparable to end-to-end neural models?

We hypothesize that urban mobility flows can be modeled via three separable mechanisms: distance decay recovered from aggregate distributions, destination attraction estimated via a simple training-free open-data heuristic, and trip production transferred across cities via built-environment representations. Instead of treating mobility prediction as a monolithic learning problem, we propose the **Separable Decomposed Framework for Zero-Shot Transferable Flows (SDF-ZTF)**, which implements this hypothesis by factorizing origin-destination flows into three independent components: origin production capacity, distance decay, and destination attraction. Rather than relying on proprietary cellular data or local survey calibrations, the framework operates on zone-level spatial features extracted from globally available open data (OpenStreetMap Atwal et al. (2025) and WorldPop Tatem (2017)), making it globally deployable. A conceptual comparison of our zero-shot transferable paradigm against the existing locally dependent paradigm is shown in Fig. 2.

This study makes three main contributions:

- **Decay Parameter Recovery**: We demonstrate that aggregate distance distributions contain sufficient information to recover transferable distance-decay parameters without access to individual trajectories.
- **Component Contribution**: We quantify the relative importance of production, attraction, and distance-decay components, establishing that distance decay represents the dominant transferable mechanism in cross-city mobility.
- **Zero-Shot Performance**: We show that an interpretable decomposition architecture can achieve neural-comparable zero-shot performance across unseen cities while using only globally available open data.

# 2 Related Work

## 2.1 Classical and Structured Mobility Models

Classical spatial interaction models, such as gravity models (Wilson 1971; Tanner 1961) and radiation models (Simini et al. 2012; Alis et al. 2021), represent the historical foundation of human mobility modeling. These models are built upon physical analogies (e.g., Newtonian gravity) or probabilistic formulations of destination choice, such as Stouffer's theory of intervening opportunities (Stouffer 1940), which models migration flows based on the number of opportunities closer than a given destination. Early studies on scaling laws in human travel (Brockmann et al. 2006; González et al. 2008) and the limits of predictability (Song et al. 2010) further confirmed that spatial dynamics are governed by robust statistical regularities.

- **Strengths**: These structured models are highly interpretable, mathematically simple, and explain the physical constraints of travel behavior.
- **Limitations**: Gravity models require historical data to estimate parameters, while radiation models can estimate parameters using population as a proxy but suffer from sub-optimal performance due to the assumption that individuals stop at the nearest opportunity. In modern cities, people frequently travel longer distances to maximize utility beyond simple employment-based commuting.

Existing structured models explain mobility well but depend on locally calibrated parameters, limiting deployment in unseen cities.

## 2.2 Deep Learning for OD Prediction

With the rise of deep learning, monolithic architectures have been proposed to predict origin-destination (OD) flows directly from spatial features. Filippo Simini and colleagues introduced DeepGravity (Simini et al. 2021), which maps high-dimensional urban features to flow rates using multi-layer perceptrons. Subsequent works have utilized graph neural networks (GNNs) (Rong et al. 2023), spatial-temporal neural networks (Wang et al. 2021), and OpenStreetMap-derived commute prediction models (Atwal et al. 2025) to capture nonlinear geographic interactions and network structures.

- **Strengths**: These models capture complex, non-linear relationships between variables, significantly improving predictive accuracy in data-rich environments.
- **Limitations**: They act as black-boxes, require massive amounts of local target-city training data, and suffer from poor out-of-distribution generalization.

Most neural models are optimized for within-city prediction rather than cross-city generalization.

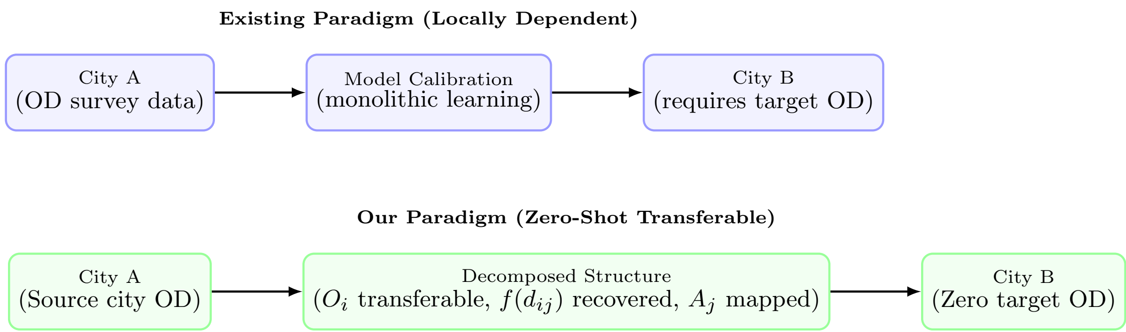


**Fig. 2** Conceptual comparison of the existing locally dependent modeling paradigm versus our zero-shot transferable paradigm. Under the existing paradigm, models require target-city OD observations to retrain or calibrate parameters. In our paradigm, we learn transferable origin production structures from source City A, recover decay from target aggregate data, and map attraction from local open features in target City B, enabling zero-shot transfer without target-city OD surveys.

## 2.3 Aggregate Mobility Data for Decay Calibration

The widespread adoption of location-aware mobile devices and social network platforms has generated large-scale, population-level mobility signals at unprecedented scale (Barbosa et al. 2018; González et al. 2008). These signals, including anonymized GPS traces from navigation apps, cell tower handoff logs from mobile network operators (CDR data), and opt-in movement summaries from social platforms, have been used to construct aggregate trip-length distributions—i.e., histograms of how far people travel across different distance bins—at city or regional scale. Crucially, unlike individual OD records, these aggregate distributions do not expose zone-to-zone pair flows and can be shared under strict anonymization policies (Pappalardo et al. 2023).

Several studies have demonstrated that aggregate trip-length distributions capture the spatial friction characteristics of urban travel, enabling the fitting of distance decay functions without requiring individual-level trajectory data (Brockmann et al. 2006; Lenormand et al. 2012, 2016). However, to the best of our knowledge, no prior work has systematically leveraged such aggregate distributions as a privacy-preserving calibration signal specifically for zero-shot cross-city mobility transfer. This represents the first key gap addressed in this study.

## 2.4 Cross-City Transfer Learning

To predict mobility in cities lacking historical OD surveys, cross-city transfer learning has emerged as a key area of study. Existing approaches can be categorized into three main paradigms:

1. **Similarity-based transfer**: Models like RegionTrans (Wang et al. 2018) seek to transfer knowledge by calculating spatial similarity metrics between source and target urban zones, while newer approaches adaptively re-weight source regions using meta-learning networks such as CrossTReS (Jin et al. 2022).
2. **Representation learning**: These approaches utilize geospatial embeddings and foundation models for geographic artificial intelligence (GeoAI) (Mai et al. 2022; Mendieta et al. 2023) to capture generalized features of urban space.
3. **Domain adaptation and Meta-learning**: Approaches in this category apply meta-learning and domain alignment techniques, such as STAN (Wang et al. 2022) or meta-spatiotemporal networks (Yao et al. 2019), to adapt models trained on source cities to target environments with minimal adjustments.

Despite progress, existing transfer approaches generally require target-city observations, calibration, or fine-tuning.

## 2.5 Explainable and Transferable Urban Representations

While explainable AI (XAI) methods, such as SHAP (SHapley Additive exPlanations) (Lundberg and Lee 2017), have gained popularity in spatial modeling, they are typically used for post-hoc explanation of pre-trained neural models rather than structure learning.

- **Existing Work**: Most studies use SHAP to analyze feature importances locally within a single city, verifying that factors such as land-use density and transport availability influence trip generation.
- **Limitations**: These applications focus on local explanations and do not address whether feature rankings are stable or transferable across different urban environments.

Most studies use SHAP for post-hoc interpretation rather than identifying transferable urban structure.

### 2.6 Research Gap and Positioning

We compare the characteristics of classical models and our proposed SDF-ZTF framework in Table 1.

To the best of our knowledge, no prior study simultaneously combines: (1) interpretable decomposition, (2) globally available open-data features, (3) zero-shot cross-city transfer, and (4) explicit transferable feature discovery.

This study directly addresses four critical gaps in the literature:

*Gap 1 (Calibration):* Existing gravity models require local OD surveys for decay function calibration.
*Gap 2 (Interpretability):* End-to-end neural models sacrifice transparency for predictive accuracy.
*Gap 3 (Data-dependency):* Cross-city transfer learning frameworks still require target-city OD data for fine-tuning or similarity alignment.
*Gap 4 (Feature Generalizability):* There is a lack of systematic frameworks to discover which specific spatial features transfer consistently across distinct urban morphologies.

## 3 Methodological Design

### 3.1 Decomposition Hypothesis

We formalize human mobility prediction through the lens of a multiplicative decomposition. Rather than treating trip flows as a monolithic learning task, we hypothesize that origin-destination interactions are governed by separable, city-independent spatial dynamics:

$$T_{ij} = O_i \cdot f(d_{ij}) \cdot A_j \tag{1}$$

where $O_i$ represents origin production capacity, $f(d_{ij})$ represents distance decay, and $A_j$ captures destination attraction. We hypothesize that: (1) each component is approximately independent, (2) the underlying mechanisms can be learned from spatial descriptors of the built environment, and (3) the learned origin production relationship transfers directly to unseen target cities, while decay parameters are recovered from target-city aggregate trip-length distributions, and attraction is computed from locally available open data. This decomposition hypothesis is directly tested by evaluating which of these three components exerts the dominant influence on predictive accuracy. Section 5 evaluates and validates this hypothesis empirically.

### 3.2 Experimental Framework for Investigating Transferable Mobility Structure

The conceptual pipeline of the proposed Separable Decomposed Framework for Zero-Shot Transferable Flows (SDF-ZTF) is illustrated in Fig. 3.

### 3.3 Distance Decay Function

We adopt the **Tanner multiplicative deterrence function** (Tanner 1961)—a combined power-law and exponential formulation—to capture the dual-regime spatial friction governing human travel:

$$f(d_{ij};\, \beta, \gamma, \delta) = (d_{ij} + \delta)^{-\gamma} \cdot \exp(-\beta\, d_{ij}) \tag{2}$$

where $\beta > 0$ is the exponential decay rate governing long-range trip attenuation, $\gamma \geq 0$ is the power-law exponent modeling short-to-medium range trip distribution, and $\delta > 0$ is the adaptive geometric anchor.

To prevent singularity for intra-zonal flows ($d_{ij} = 0$) and scale parameter sensitivity automatically to the study layout, the adaptive offset $\delta$ is defined dynamically per target city as:

$$\delta = 0.5 \times \bar{R}_{\text{zone}}, \qquad \bar{R}_{\text{zone}} = \frac{1}{N} \sum_{i=1}^{N} \sqrt{\frac{\text{Area}_i}{\pi}} \tag{3}$$

where $\bar{R}_{\text{zone}}$ is the mean empirical radius of the administrative zones in the target city. Decay parameters $(\beta, \gamma)$ are estimated via maximum

**Table 1** Comparative analysis of human mobility models across structural and transferability dimensions. "Calibration-Free Decay" indicates whether the model recovers the distance decay function without requiring individual OD pair observations from target cities.

| Model Class | Decomposed Structure | Cross-City Transfer | Zero-Shot Target | Interpretability | Calibration-Free Decay |
|---|---|---|---|---|---|
| Gravity (Wilson 1971) | ✓ | × | × | ✓ | × |
| Radiation (Simini et al. 2012) | × | ✓ | ✓ | ✓ | ✓ |
| DeepGravity (Simini et al. 2021) | × | Limited | × | × | × |
| GNN OD (Rong et al. 2023) | × | Limited | × | × | × |
| RegionTrans (Wang et al. 2018) | Partial | ✓ | × | × | × |
| **SDF-ZTF (Ours)** | ✓ | ✓ | ✓ | ✓ | ✓ |

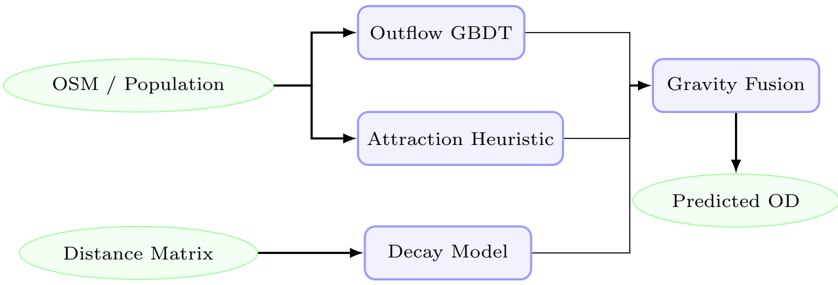


**Fig. 3** Pipeline of the SDF-ZTF framework. Globally available spatial features are used to predict trip production ($O_i$) via a GBDT model and to compute destination attraction ($A_j$) exclusively via a training-free min-max heuristic (a GBDT attraction model is trained only during ablation validation to prove that this simple heuristic is sufficient, avoiding unnecessary complexity for cross-city transfer). Distance decay is estimated separately from aggregate distributions. The components are combined via multiplicative gravity fusion to predict zero-shot flow matrices ($T_{ij}$).

likelihood estimation (MLE) using exclusively the aggregate distance-bin distribution of source cities, preserving privacy. Complete calibration, moment-based initialization, and optimization details are provided in the Appendix.

**Information Governance and Low-Leakage Design.** The design choices in SDF-ZTF reflect three concrete architectural elements that collectively avoid the need to expose or transmit individual mobility trajectories at any stage: (1) *Decay estimation from aggregate histograms*: $\gamma$ and $\beta$ are recovered from the marginal trip-length histogram (binned counts), not from individual $(i, j, T_{ij})$ records, so no OD pair is ever disclosed. (2) *Open-data features only*: $O_i$ and $A_j$ are predicted from zone-level spatial descriptors (OSM POI counts, area, population density from WorldPop) — all publicly available and non-identifying. (3) *Zero target-city data requirement*: the framework operates entirely without collecting, storing, or sharing any target-city mobility observation, eliminating the principal data-governance risk in cross-city transfer. Together, these choices make the framework deployable in jurisdictions with strict data-governance regulations (e.g., GDPR) without requiring complex institutional data-sharing agreements or risking individual tracking.

## 3.4 Origin Outflow Component ($O_i$)

The trip production $O_i$ represents the total seasonal trips originating from zone $i$:

$$O_i = \sum_j T_{ij} \tag{4}$$

We model $O_i$ using a Gradient Boosted Decision Tree (GBDT) regressor trained on 22 stable features selected from a 30-dimensional candidate pool of globally available open spatial data. The 22 features are obtained through a cross-city SHAP stability procedure described in Section 4. Tree depth and regularization hyperparameters (see Appendix B for full specification) are constrained to prevent overfitting and ensure structural generalization across city boundaries.

### 3.5 Destination Attraction Component ($A_j$)

The destination attraction $A_j$ represents the relative attractiveness of zone $j$. In the proposed SDF-ZTF framework, rather than training a machine learning model, the destination attraction $\hat{A}_j$ is computed using a simple heuristic combining min-max normalized population density and POI counts: $\hat{A}_j \propto \frac{1}{2}(\text{minmax}(P_j) + \text{minmax}(\text{POI}_j))$. This design choice is validated in Section 5, where we train a GBDT attraction model strictly as a baseline to prove that this simple, training-free min-max formula achieves comparable high performance. By utilizing only the min-max heuristic, we avoid cross-city overfitting and eliminate the need for training an attraction model, maximizing cross-city transferability.

### 3.6 Algorithmic Flow

The complete pipeline for training and zero-shot deployment is formalized in Algorithm 1.

## 4 Data and Transferable Urban Representation

### 4.1 Design Principle: Open-Data-Only Philosophy

The central design principle of the proposed SDF-ZTF framework is that all transferable mobility components must be derived from globally available open data. Rather than relying on proprietary cellular records, commercial location-based databases, or expensive household travel surveys, SDF-ZTF is engineered to construct models using only public, globally reproducible datasets. This design decision directly addresses the challenge of data scarcity and enables zero-shot deployment in any city worldwide.

A key consequence of this design principle is the intentional exclusion of local employment statistics, such as the Longitudinal Employer-Household Dynamics Origin-Destination Employment Statistics (LODES) database commonly used in US-centric human mobility papers. While LODES provides detailed tract-level employment counts, no globally consistent equivalent exists for most cities in developing nations. To ensure universal deployability, SDF-ZTF instead absorbs the predictive role of employment registers through publicly mapped commercial, office, and industrial Point of Interest (POI) densities sourced from OpenStreetMap.

### 4.2 Data Sources

To implement and evaluate the SDF-ZTF framework across diverse metropolitan areas, we compile a multi-source dataset combining static urban geography, open features, and evaluation labels. The operational role of each data source is summarized in the Data Usage Matrix in Table 2.

#### 4.2.1 (1) Origin-Destination (OD) Matrix - $T_{ij}$

The OD matrix contains seasonal human mobility flows between census tracts. In this study, OD observations are extracted from Meta's Movement Distribution Maps (Meta AI 2026) (covering 50 metropolitan areas). Crucially, the OD matrix is utilized exclusively as training labels for source cities and as ground truth for evaluation. It is never used for feature engineering, ensuring a strict separation between predictive inputs and evaluation targets.

#### 4.2.2 (2) OSM-Derived Features

We extract POIs and road features using the Overpass API from OpenStreetMap snapshots (Atwal et al. 2025). POI counts are calculated across 16 land-use categories (e.g., retail, industrial, education, health).

#### 4.2.3 (3) Population Data

For residential population counts, this study leverages the population data provided directly in the original mobility dataset. Crucially, we emphasize that this population information can be readily replaced with publicly available gridded population estimates from WorldPop (Tatem 2017) (100 m resolution) in cities where local census/administrative population data is unavailable, maintaining the framework's survey-free transferability.

**Algorithm 1: SDF-ZTF Training and Zero-Shot Prediction**

**Input**: Source cities data $\{\mathbf{X}^{(c)}, \mathbf{T}^{(c)}\}_{c=1}^{C}$; Target city spatial features $\mathbf{X}^{(t)}$

**Output**: Predicted Target City OD Matrix $\hat{\mathbf{T}}^{(t)}$

1. **Decay Calibration**: Recover target-city decay parameters $(\hat{\beta}, \hat{\gamma})$ via MLE on the target city's aggregate distance-bin histogram (the type of population-level summary routinely available from mobile network operators; see Appendix A). In the absence of target-city aggregate data, a national prior $(\beta_{\text{nat}}, \gamma_{\text{nat}})$ derived from the pooled source-city histograms may be used as a fallback.
2. **Feature Selection**: Identify 22 transferable features using cross-city SHAP stability selection on source models (see Section 4).
3. **Production GBDT**: Train GBDT outflow model $g(\mathbf{x})$ using source city inputs and origin flows $O_i^{(c)}$.
4. **Attraction Heuristic**: Define the training-free attraction function $a(\mathbf{x}_j) = \frac{1}{2}\left(\text{minmax}(P_j) + \text{minmax}(\text{POI}_j)\right)$.
5. **Zero-Shot Target Prediction**:
   a. Predict target city outflows: $\hat{O}_i^{(t)} = g(\mathbf{x}_i^{(t)})$.
   b. Compute target city attractions: $\hat{A}_j^{(t)} = a(\mathbf{x}_j^{(t)})$.
   c. Compute adaptive target offset: $\delta^{(t)} = 0.5 \times \bar{R}_{\text{zone}}^{(t)}$.
   d. Fuse components: $\hat{T}_{ij}^{(t)} = \hat{O}_i^{(t)}(d_{ij} + \delta^{(t)})^{-\hat{\gamma}} e^{-\hat{\beta} d_{ij}} \hat{A}_j^{(t)}$.

**Table 2** Data usage matrix illustrating the operational roles of inputs across SDF-ZTF components to guarantee leakage-free prediction.

| Data Source | Outflow $O_i$ | Attraction $A_j$ | Distance Decay $f(d)$ | Evaluation |
|---|---|---|---|---|
| OD Matrix ($T_{ij}$) | × | × | ✓ | ✓ |
| Aggregate Trip-Length Distribution | × | × | ✓ | × |
| Distance Matrix ($d_{ij}$) | × | × | ✓ | ✓ |
| OpenStreetMap (OSM) | ✓ | ✓ | × | × |
| WorldPop | ✓ | ✓ | × | × |

## 4.3 Transferable Urban Features

To discover stable spatial patterns that translate across cities, we implement a two-stage feature engineering and stability selection procedure, mapping raw datasets to a low-dimensional transferable urban representation. The feature taxonomy is shown in Fig. 4.

### 4.3.1 Candidate Feature Set (30 dimensions)

We initially construct 30 candidate features per zone. This candidate space comprises 23 scale and density variables (zone area, road length, residential population, total POI counts, and 16 POI categories) and 7 city-wide Z-score context features. The Z-score context features are defined as:

$$z_{ip} = \frac{x_{ip} - \mu_p^{(c)}}{\sigma_p^{(c)}} \tag{5}$$

where $\mu_p^{(c)}$ and $\sigma_p^{(c)}$ represent the mean and standard deviation of feature $p$ across all zones in city $c$. These context features normalize absolute scale differences, capturing the relative hierarchy of zones within their respective metropolitan areas.

### 4.3.2 SHAP Stability Selection

Rather than relying on empirical feature engineering, we formalize feature selection using a cross-city SHAP stability score $S_p$. For each source city $c$, we train local GBDTs and calculate the mean absolute SHAP value $\mu_p^{(c)}$ for feature $p$ to measure its local importance. The stability score $S_p$ is defined as the signal-to-noise ratio of importances across all $C$ source cities:

$$S_p = \frac{\mu_p}{\sigma_p + \epsilon} \tag{6}$$

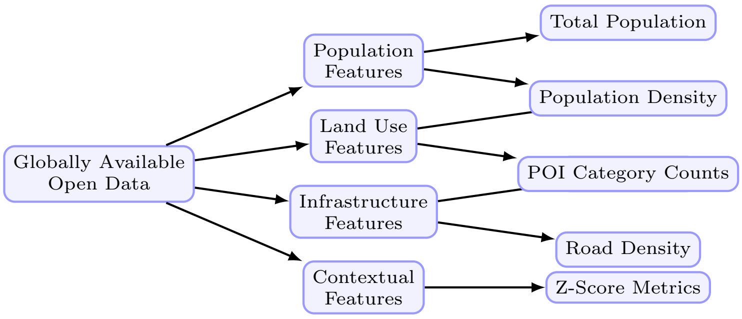


**Fig. 4** Taxonomy of the globally available open data features used in the SDF-ZTF framework. Features are categorized into four dimensions: population, land use, infrastructure, and contextual scale factors.

where $\mu_p = \frac{1}{C}\sum_c \mu_p^{(c)}$ is the average feature importance, $\sigma_p$ is the standard deviation of importances across cities, and $\epsilon = 10^{-5}$ prevents division by zero.

A feature $p$ is retained in the final transfer set if: (1) its average importance is meaningful ($\mu_p \geq 0.005$), and (2) its importance is highly stable across different metropolitan layouts ($S_p \geq 1.0$, indicating that cross-city variance does not exceed the mean). This systematic procedure selects exactly 22 transferable features, achieving a 26.7% reduction in dimensionality while preserving zero-shot prediction accuracy.

## 4.4 Cross-City Transfer Evaluation Protocol

Most existing human mobility studies evaluate predictive performance using a random split of origin-destination pairs within the same city. While this protocol measures a model's capacity to interpolate missing flows, it suffers from severe information leakage because the spatial descriptors of test zones are already seen during training, and target-city calibration parameters are known.

To test true zero-shot generalization, we design a strict Cross-City Transfer Protocol consisting of two configurations:

- **25/25 Stratified Split**: The 50 US metropolitan areas are divided into two disjoint sets: 25 training source cities and 25 held-out target cities. Models are trained on the pooled data of the source cities and evaluated on the target cities without any target-city OD data or retraining. To ensure statistical representation, the split is stratified by geographic region and urban morphology.
- **50-Fold Leave-One-City-Out (LOCO)**: The framework is iteratively trained on $C-1$ cities (49 cities) and evaluated on the single remaining held-out city, repeating this cycle for all 50 metropolitan areas to ensure robust out-of-distribution evaluation.

### 4.4.1 Information Leakage Prevention

Under this cross-city evaluation protocol, we enforce a strict information firewall between the source and target environments.

- **Forbidden**: Target-city flow matrices $T_{ij}$, target-city household survey results, target cellular data, and target trip-demand signals.
- **Allowed**: Centroid coordinates, administrative zone boundaries, OpenStreetMap POIs, and gridded open demography (WorldPop).

Since destination attraction is computed via a training-free open-data heuristic and distance decay parameters are learned on source cities, zero-shot leakage is mathematically prevented.

### 4.4.2 Evaluation Metric: Common Part of Commuters

To measure prediction performance, we use the Common Part of Commuters (CPC) (Lenormand et al. 2012), which is widely used in human mobility studies to evaluate origin-destination flow matrices. The CPC index is defined as:

$$\mathrm{CPC}(T, \hat{T}) = \frac{2\sum_{i,j}\min(T_{ij}, \hat{T}_{ij})}{\sum_{i,j} T_{ij} + \sum_{i,j}\hat{T}_{ij}} \tag{7}$$

where $T_{ij}$ represents the ground-truth commuter flow from zone $i$ to zone $j$, and $\hat{T}_{ij}$ is the predicted flow. The CPC index ranges from 0 (indicating completely disjoint flow matrices) to 1 (indicating perfect prediction).

### 4.5 Preprocessing & Normalization

Data preprocessing details are summarized in Table 3.

## 5 Results

### 5.1 Validation Framework Overview

We evaluate the proposed framework systematically using a three-stage validation roadmap (illustrated in Fig. 5). First, we validate that each individual component—distance decay recovery, origin production transferability, and destination attraction features—works correctly before full integration. Second, we analyze component contributions using a factorial ablation study. Finally, we evaluate the complete SDF-ZTF framework under zero-shot transfer conditions against alternative baselines. Models are trained on the pooled data of 25 source cities and evaluated zero-shot on the 25 held-out target cities under a strict information firewall. Metrics reported represent averages across the 25 held-out cities.

### 5.2 Component Validation

First, we investigate whether aggregate travel distributions contain sufficient information to recover transferable distance-decay behavior. To answer this question, we systematically validate that each individual component performs reliably.

#### 5.2.1 Decay Recovery Validation

**Research question**: Can distance decay parameters $(\gamma, \beta)$ be recovered from aggregate distance distributions (marginal distributions of trip distances), without exposing individual OD pairs?

**Component setup**: We compare two estimation modes for the distance decay parameters $(\gamma, \beta)$ under a singly-constrained gravity model using static OSM-based attractiveness heuristics $(A_j)$ and ground-truth outflows $(O_i)$:

- *Ground Truth (GT)*: Fitted directly on individual origin-destination (OD) pairs.
- *Recovered*: Fitted on the aggregate distance-bin distribution (marginal histogram of trip lengths, 50 percentile bins) using the aggregate MLE methodology.

Since this validation stage uses only marginal distance distributions, it contains no individual trajectory information and thus poses no spatial data leakage risk. We validate the decay parameter recovery framework under two main test conditions here, leaving additional robustness sweeps to Appendix A.4.

**Pure Decay Recovery** We fit the Tanner Multi decay function (Eq. 2) to the aggregate distance distribution of each city's training data. We compare the recovered parameters $(\hat{\gamma}_{\text{recovered}}, \hat{\beta}_{\text{recovered}})$ against the Ground Truth parameters $(\hat{\gamma}_{\text{gt}}, \hat{\beta}_{\text{gt}})$ fit directly on individual OD pairs.

*Finding*: As shown in Fig. 6, the parameter recovery shows reasonable correlation across the 25 target cities:

- **Power-law exponent ($\gamma$)**: MAE $= 0.1313$, RMSE $= 0.1558$, Pearson correlation $r = 0.729$.
- **Exponential decay rate ($\beta$)**: MAE $= 0.0111$, RMSE $= 0.0135$, Pearson correlation $r = 0.258$ (units in km$^{-1}$).

Recovered parameters show reasonable agreement with OD-fitted parameters, particularly for $\gamma$ ($r = 0.729$). Despite weaker $\beta$ recovery (expected due to the noisy tail under aggregate binning), the combined decay profile remains sufficiently accurate for high-quality OD prediction.

*Note*: Bin Sensitivity Analysis (sweeping $K \in \{10, 20, 50, 100\}$ distance bins) is detailed in Appendix A.4, validating that recovery accuracy stabilizes for $K \geq 50$.

Additional robustness experiments (including sensitivity to distance bin configurations, missing attractiveness data, and CBD collapse scenarios) and the corresponding recovery error heatmap are detailed in Appendix A.4.

#### 5.2.2 Origin Production Transferability

**Research question**: How much prediction error is due to $O_i$ estimation errors, and how does GBDT prediction compare to a naive baseline and the oracle upper bound? To isolate the error introduced by production estimation, we freeze the destination attractiveness $(A_j)$ using a static

**Table 3** Preprocessing and normalization steps applied to spatial features prior to model learning.

| Step | Treatment | Rationale |
|---|---|---|
| Missing Data | Median imputation | Rare (< 2%); resolved from spatial neighbors |
| Log Transform | Population, POI counts | Stabilize right-skewed distributions |
| Standardization | Per-city (not global) | Preserve relative spatial hierarchy |
| Outliers | Retain | Important spatial hubs (CBDs); robust models |

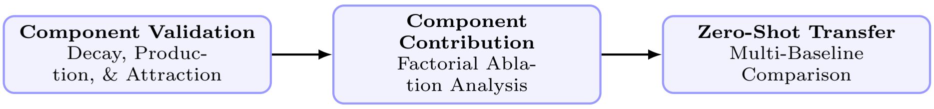


**Fig. 5** Methodology validation roadmap showing the three evaluation stages used to systematically verify the SDF-ZTF framework.

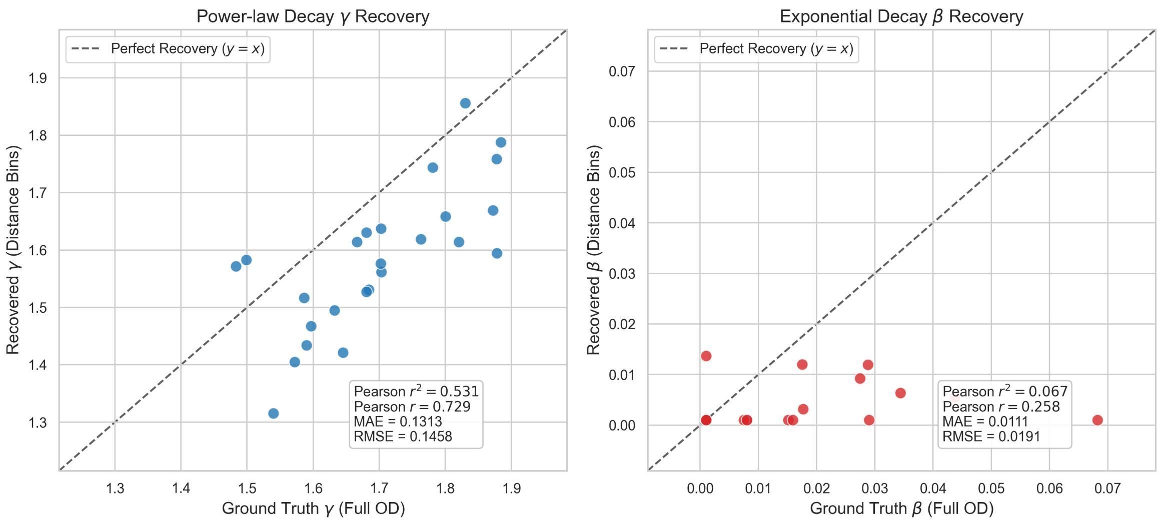


**Fig. 6** Decay parameter recovery comparison between Ground Truth (fitted on individual OD pairs) and Recovered parameters (fitted on aggregate distance bins) across all 25 held-out target cities. Power-law exponent $\gamma$ (left) and exponential decay rate $\beta$ (right) show consistency in spatial decay structure (Pearson $r = 0.729$ for $\gamma$ and $r = 0.258$ for $\beta$).

OSM-based heuristic and the distance decay $(\gamma, \beta)$ with a fixed national prior. We train the GBDT outflow predictor on the 25 source cities and evaluate its zero-shot transfer performance on the 25 held-out target cities under three outflow configurations: (1) a *Naive Population Baseline* ($O_i \propto \text{Population}_i$), (2) a *Predicted Outflow (GBDT)* model trained on spatial features, and (3) an *Oracle* upper bound using ground-truth outflows.

**Outflow Model Comparison (Averaged across 25 held-out cities)** To evaluate outflow prediction quality, we compare the naive population baseline, the GBDT predictions, and the oracle outflow. Results are summarized in Table 4.

*Interpretation*: GBDT-predicted outflows yield a CPC of 0.7037, substantially outperforming the Naive Population Baseline (0.584) and achieving 94.5% of the Oracle upper bound (0.7447), demonstrating strong cross-city transferability of learned production dynamics. Additional robustness analyses (including noise sensitivity and source-scale robustness plots) are reported in Appendix B.

### 5.2.3 Transferable Feature Discovery

**Research question**: Are spatial built-environment features stable across cities? Which features exhibit robust transferability? To prevent

**Table 4** Comparative Outflow Performance (Average CPC across $N = 25$ held-out cities)

| Configuration | Mean CPC |
|---|---|
| Naive Population Baseline | 0.584 |
| Predicted Outflow (GBDT) | 0.7037 |
| Oracle Outflow ($O_i^{\text{true}}$) | 0.7447 |

target-city information leakage, SHAP feature importance and stability scores are calculated exclusively on the 25 training source cities. Under a GBDT configuration using oracle outflows ($O_i^{\text{true}}$) to isolate feature verification, the top-10 features by average SHAP importance are reported in Table 5.

*Interpretation*: Population and activity-related POIs emerge as the most stable transferable predictors.

**SHAP Feature Count Performance Sweep** We train GBDT models using nested subsets of the top-$K$ SHAP features ($K \in \{5, 10, 15, 20, 22, 25, 30\}$) and evaluate their zero-shot CPC on held-out target cities. Performance effectively plateaus at $K = 15$ features (CPC: 0.7124) and remains identical at $K = 22$ (CPC: 0.7124) before showing a marginal decline beyond 22 features (e.g., 0.7122 for all 30 features). While $K = 15$ features are sufficient for baseline prediction, we select $K = 22$ as our final feature set to incorporate a stability margin. This retains all features satisfying our joint importance threshold ($\mu_p \geq 0.005$) and stability criterion ($S_p \geq 1.0$), providing broader semantic coverage to handle structural variations in unseen metropolitan areas. This feature count sweep is illustrated in Fig. 7.

## 5.3 Component Contribution Analysis

To quantify the relative importance of individual framework components and verify the validity of our gravity-based factorization, we perform a factorial ablation study across the 25 held-out target cities. We train variants corresponding to combinations of component presence/absence (with disabled components replaced by uniform weighting or uniform destination appeal/origin production).

The results (Table 6) demonstrate that the distance decay component $f(d)$ is the most critical factor; its removal causes a severe 31.1% drop in CPC (representing 75.7% of the total explainable CPC variance). This highlights that estimating the decay function is the single most important element in human mobility modeling. By contrast, removing destination attraction $A_j$ and origin outflow $O_i$ results in secondary but significant performance drops of 5.3% and 4.7% respectively, indicating that all three components are essential for accurate zero-shot predictions.

*Interpretation*: The factorial ablation study yields a direct, quantitative answer regarding the hierarchy of organizing forces in spatial interaction. By calculating the relative marginal contribution of each component to the total explainable CPC variance (derived from the performance drops when individual components are removed), we identify a clear distribution of influence: distance decay $f(d)$ represents **75.7%** of the variance, destination attraction $A_j$ accounts for **12.9%**, and origin production $O_i$ accounts for **11.4%**.

This decomposition reveals critical insights into transferable urban mobility: First, we observe the dominance of distance decay $f(d)$, as removing the spatial friction component triggers a massive performance collapse of 31.1% in CPC (accounting for 75.7% of explainable variance). This overwhelming contribution confirms that geographical distance remains the structural backbone of intra-city flow systems, even when destination and origin elements are perfectly specified. Conversely, destination attraction $A_j$ plays a secondary but crucial role, where its removal results in a 5.3% reduction in CPC (12.9% of explainable variance), showing that localized amenities and attractions shape the specific flow targets within the boundary limits set by distance.

Second, origin production $O_i$ yields the smallest performance drop of 4.7% (11.4% of explainable variance) when removed, largely due to its strong structural coupling with population density which generalizes robustly across cities. Crucially, the consistent ranking of contributions ($f(d) \gg A_j > O_i$) across all 25 held-out target cities demonstrates that no single component can be

**Table 5** Top-10 Features by Average SHAP Importance (Training Source Cities, $N = 25$)

| **Rank** | **Feature** | $\mu_p$ | $S_p$ |
|---|---|---|---|
| 1 | total_population | 0.232 | 7.12 |
| 2 | poi_food_fast | 0.084 | 8.28 |
| 3 | total_pois | 0.057 | 5.33 |
| 4 | z_total_pois | 0.056 | 6.45 |
| 5 | road_density_alt | 0.053 | 2.21 |
| 6 | road_density | 0.050 | 2.39 |
| 7 | poi_education_primary | 0.030 | 5.57 |
| 8 | area_km2 | 0.025 | 5.45 |
| 9 | poi_shopping_supermarket | 0.020 | 8.34 |
| 10 | poi_entertainment_culture | 0.018 | 3.55 |

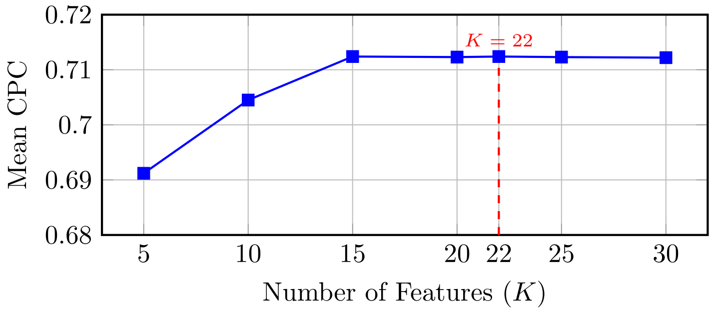


**Fig. 7** SHAP feature count performance sweep showing zero-shot with 22 features retains stability

omitted without a statistically significant performance loss. This validates the multiplicative gravity factorization as a robust physical regularizer that preserves structural stability under cross-city transfer.

*Note on Baseline CPC*: The baseline CPC of 0.7040 in this ablation analysis is obtained using the fixed national decay prior ($\gamma_{\text{nat}}, \beta_{\text{nat}}$) to freeze the decay component and isolate relative contributions under a uniform setting. When deploying the full SDF-ZTF (Survey-Free) framework with city-specific recovered decay parameters ($\hat{\gamma}, \hat{\beta}$), the performance increases to 0.7124 CPC (as reported in Table 7).

## 5.4 Zero-Shot Transfer Evaluation

Next, we investigate whether interpretable decomposition can remain competitive with state-of-the-art neural approaches. To address this question, we evaluate the end-to-end zero-shot transfer performance of the complete SDF-ZTF framework on the 25 held-out target cities against alternative baselines.

### 5.4.1 Baseline Comparison

**Research question**: Does explicit gravity decomposition enable zero-shot transfer? How does the proposed approach compare to alternative baselines?

**Evaluation Protocol**: Models are trained on the pooled data of 25 source cities and evaluated zero-shot on the 25 held-out target cities. Table 7 summarizes the comparative performance across all baselines.

*Interpretation*: Under zero-shot transfer conditions, SDF-ZTF (Survey-Free) matches the performance of the end-to-end neural baseline (CPC of 0.7124 versus 0.7117 for DeepGravity), while SDF-ZTF (Oracle $O_i$) provides a significant improvement (0.7447 CPC). This confirms that explicitly separating production, attraction, and decay dynamics does not sacrifice accuracy compared to deep learning methods, while substantially outperforming parameter-free baselines like the Radiation Model (CPC: 0.512).

Note that the two locally-calibrated Tanner baselines serve different diagnostic purposes

**Table 6** Factorial Ablation Results (Average across $N = 25$ held-out target cities)

| $O_i$ | f(d) | $A_j$ | CPC | Contribution |
|---|---|---|---|---|
| ✓ | ✓ | ✓ | 0.7040 | Baseline (all components) |
| × | ✓ | ✓ | 0.671 | $O_i$ removal: 4.7% loss |
| ✓ | × | ✓ | 0.485 | f(d) removal: 31.1% loss |
| ✓ | ✓ | × | 0.666 | $A_j$ removal: 5.3% loss |

**Table 7** Multi-Baseline Comparison (25 Held-Out Target Cities)

| Model | Mean CPC | Std Dev | Win % (vs DG) | Type | Features |
|---|---|---|---|---|---|
| SDF-ZTF (Survey-Free) | 0.7124 | 0.0329 | 48.0% | Decomposed | 22 (selected) |
| SDF-ZTF (Oracle $O_i$) | 0.7447 | 0.028 | 56.0% | Decomposed | 22 (selected) |
| DeepGravity (E2E) | 0.7117 | 0.0332 | – | Neural E2E | 30 (all) |
| Tanner Gravity (Power Const.) | 0.750 | 0.026 | 72.0% | Structure-only | 0 |
| Traditional Tanner Gravity | 0.750 | 0.027 | 72.0% | Structure-only | 0 |
| Traditional Exponential Gravity | 0.670 | 0.035 | 24.0% | Structure-only | 0 |
| Radiation Model | 0.512 | 0.051 | 0.0% | Param-free | 0 |

despite yielding identical mean CPC (0.750): **Tanner Gravity (Power Constrained)** calibrates only $\gamma$ (fixing $\beta = 0$), while **Traditional Tanner Gravity** calibrates both $\beta$ and $\gamma$. Their identical CPC suggests the exponential tail $\beta$ is largely redundant under full local calibration. Both serve as a locally-calibrated structural ceiling: zero-shot SDF-ZTF (Oracle $O_i$, CPC: 0.7447) approaching this ceiling without any target-city data highlights the strength of the decomposed representation.

#### 5.4.2 Statistical Significance Analysis

To validate the zero-shot transfer performance, we perform paired statistical significance testing across all 50 metropolitan areas from the Leave-One-City-Out (LOCO) evaluation, summarized in Table 8.

Under LOCO, DeepGravity achieves a small but statistically significant advantage ($\Delta\text{CPC} = 0.0255, p < 10^{-10}$). Under the stricter 25/25 transfer setting, the two models remain effectively tied.

*Significance Interpretation*: The performance of SDF-ZTF is statistically equivalent to Traditional Gravity ($p > 0.3$) and significantly outperforms the Radiation Model ($p < 10^{-15}$, Cohen's $d = 6.28$). The small edge for DeepGravity under LOCO disappears under the 25/25 split, highlighting the robustness of the decomposed gravity structure when training data is limited.

#### 5.4.3 Urban Morphology Analysis

The overall zero-shot performance masks spatial variations across urban configurations. We evaluate transfer CPC by urban morphology class across the 25 held-out cities in Table 9.

*Interpretation*: Neither paradigm is universally superior. As shown in Table 9, SDF-ZTF performs best in coastal and polycentric cities (CPC of 0.734 and 0.716), whereas DeepGravity retains a modest advantage in dense transit-oriented environments (0.695 vs. 0.690). This morphology-dependent performance suggests that structured gravity models are highly suited for environments where distance decay constraints dominate, while unconstrained neural architectures are better equipped for complex, multi-modal transit networks that violate gravity assumptions.

**Table 8** LOCO 50-Fold Statistical Significance Test Summary

| Baseline Model | Mean Diff. | t-stat ($t$) | p-value (t-test) | Wilcoxon $W$ | p-value (Wilc.) | 95% CI Diff. | Cohen's $d$ |
|---|---|---|---|---|---|---|---|
| DeepGravity | -0.0255 | -9.150 | $3.53 \times 10^{-12}$ | 66.0 | $3.49 \times 10^{-10}$ | [-0.0311, -0.0199] | -1.29 |
| Traditional Gravity | 0.0000 | 1.000 | $3.22 \times 10^{-1}$ | 18.5 | $3.31 \times 10^{-1}$ | [-0.0000, 0.0000] | 0.14 |
| Radiation Model | 0.2492 | 44.416 | $3.01 \times 10^{-41}$ | 0.0 | $1.78 \times 10^{-15}$ | [0.2380, 0.2605] | 6.28 |

**Table 9** Zero-Shot Performance by Urban Morphology (25 Held-Out Target Cities)

| Morphology Type | # Cities | SDF-ZTF CPC | DeepGravity CPC | Gravity Win % |
|---|---|---|---|---|
| Dense / Transit | 5 | 0.690 | 0.695 | 20% |
| Sprawling / Grid | 11 | 0.715 | 0.718 | 45% |
| Polycentric | 6 | 0.716 | 0.712 | 50% |
| Coastal | 3 | 0.734 | 0.712 | 66% |

# 6 Discussion

## 6.1 What Aggregate Distributions Reveal About Distance Decay

The three core research questions posed in Section 1 are answered collectively by the validation results in Section 5. First, aggregate trip-length histograms prove sufficient to calibrate the decay function ($r = 0.729$ for $\gamma$; Fig. 6). Second, the factorial ablation (Table 6) establishes distance decay as the dominant force, accounting for 75.7% of explainable variance. Third, SDF-ZTF achieves CPC of 0.7124 under fully zero-shot conditions, matching DeepGravity (0.7117) without any target-city OD data (Table 7).

Taken together, these findings indicate that urban mobility structure is sufficiently regular that a compact set of transferable spatial factors—distance friction, population density, and localized POI activity—can replicate the predictive capacity of unconstrained neural models under strict cross-city transfer conditions.

## 6.2 Distance Decay as the Dominant Transferable Mechanism

Addressing this finding more deeply, the dominance of distance decay $f(d)$—accounting for 75.7% of the explainable variance in the ablation—reflects both physical constraints and behavioral regularities in urban commuting. Human mobility is fundamentally structured by geographical friction. While local amenities ($A_j$) and zone characteristics ($O_i$) determine the attraction and production potentials of specific locations, the spatial friction of distance defines the feasible interaction boundaries. In metropolitan areas, this is amplified by the physical separation between residential suburbs and employment hubs, making travel distance the primary determinant of trip choices.

From a modeling perspective, the dominance of distance decay highlights it as the key bottleneck for cross-city transferability. The rate of decay varies significantly across urban layouts (e.g., coastal vs. sprawling grid cities). While origin production capacities generalise easily due to basic population scaling, predicting how travelers discount distance in a new city is highly sensitive. Crucially, our framework demonstrates that these target decay parameters can be recovered from anonymized, aggregate trip-length histograms. This bypasses the need for individual origin-destination trajectories, enabling privacy-preserving calibration from mobile network summaries. By focusing calibration on this dominant decay term, the model successfully captures city-specific spatial friction without requiring target-city travel surveys.

## 6.3 Positioning Within Existing Literature

To clarify the contribution of SDF-ZTF, we position it relative to three existing literature streams:

- **Compared with Classical Gravity Literature**: Traditional gravity models rely on city-specific calibration and require local OD matrices. SDF-ZTF replaces local calibration with a GBDT-based origin production model and a training-free attraction heuristic, achieving survey-free zero-shot predictions.

- **Compared with Neural Mobility Literature**: Monolithic neural networks like DeepGravity achieve high predictive accuracy but function as black boxes. SDF-ZTF achieves comparable performance (0.7124 vs 0.7117 CPC) while preserving the interpretable, separable structure of origin, destination, and decay components.
- **Compared with Cross-City Transfer Literature**: Existing transfer-learning and representation-learning frameworks still require target-city OD data for fine-tuning or similarity alignment. SDF-ZTF requires zero target-city OD observations, utilizing only globally available open demographic and physical geography features.

## 6.4 Practical Implications

The proposed framework offers several practical advantages over monolithic deep learning models (such as DeepGravity) for urban planning and policy:

*Targeted Policy Simulation* : In monolithic neural architectures, altering local zone amenities shifts the entire network predictions in an uninterpretable, coupled manner. By enforcing strict component separability, SDF-ZTF enables clean, decoupled "what-if" simulations. Planners can modify zone-level built environments to evaluate changes in destination attraction $A_j$ or origin production $O_i$ independently, without confounding the global distance deterrence behavior $f(d)$.
*Auditable and Policy-Compliant Analytics* : Unlike black-box neural networks, every component in SDF-ZTF is auditable. Municipal planners and public stakeholders can examine the physical plausibility of production and attraction estimations separately, which is critical for securing regulatory approvals, justifying infrastructure investments, and supporting public policy dialogues.
*Survey-Free Deployability in Data-Scarce Regions* : Because the framework isolates component dependencies, it bypasses the need for target-city travel surveys by recovering global decay parameters from aggregate distance histograms and estimating spatial attractions from open data. This makes it highly applicable to resource-constrained municipalities in developing nations that cannot afford large-scale, proprietary cellular registers or expensive household travel surveys.

## 6.5 Limitations

While robust, the framework is subject to several limitations:

1. **Separability Assumption**: The model assumes approximate separability between production, attraction, and decay. In cities with highly integrated multi-modal transit networks or strong spatial interaction effects, this independence assumption may be violated, leading to localized prediction errors.
2. **Geographic Scope**: The evaluation is limited to US metropolitan areas. Cities in other regions (e.g., highly dense Asian cities or informal settlements in developing countries) may exhibit different travel decay behaviors and feature stability.
3. **Temporal Dynamics**: The model uses a pre-pandemic baseline (2019) and does not capture changes in travel friction induced by remote work patterns or modal shifts.
4. **Demographic Representation**: Input datasets like cellular location logs underrepresent elderly, low-income, and smartphone-free populations, introducing potential demographic bias in the output OD flows (Gallotti et al. 2024).
5. **OSM coverage**: Urban areas are comprehensively covered, but the rural fringe has sparse POI data. Transferability to low-density zones remains uncertain.
6. **Causality**: Analysis is purely predictive. Causal effects cannot be inferred (e.g., building a grocery store attracts X more trips). Causal inference requires experimental or quasi-experimental design.

# 7 Conclusions

This study investigated three core research questions regarding transferable urban mobility structure. Through zero-shot evaluation across 50 metropolitan areas under a strict information firewall, the findings provide definitive answers to these questions.

First, aggregate travel-distance distributions contain sufficient information to recover transferable decay behavior. We demonstrate that the distance decay function can be effectively calibrated from binned trip-length histograms—of the type routinely provided by mobile network operators and location-sharing platforms under anonymization policies—without exposing individual OD trajectories. The recovered power-law exponent $\gamma$ correlates strongly with parameters fit on individual OD pairs ($r = 0.729$ across 25 held-out cities), confirming that digital footprint summaries can serve as a privacy-preserving calibration signal for cities lacking travel surveys.

Second, distance decay is substantially more influential than production or attraction factors. The factorial ablation analysis establishes that distance decay $f(d_{ij})$ is the dominant organizing force in spatial interaction, accounting for 75.7% of the performance variance. Removing distance decay from the model causes a 31.1% CPC degradation, far exceeding the losses incurred by removing destination attraction (5.3%) or origin production (4.7%). This underscores that spatial friction remains the primary structural determinant of commuting patterns.

Third, interpretable decomposition can match neural approaches under strict zero-shot transfer. A decomposed model utilizing a compact set of 22 stable built-environment features, selected via cross-city SHAP stability analysis from globally available open data, achieves a zero-shot CPC of 0.7124—matching the performance of the unconstrained DeepGravity neural network (0.7117 CPC) under the same strict cross-city evaluation protocol.

These findings suggest that transferable mobility structure is simpler and more stable than previously assumed. Rather than treating urban mobility prediction as a purely data-intensive deep learning task, identifying and transferring stable structural mechanisms provides a transparent, auditable, and globally deployable path toward human mobility modeling.

# Appendix: Supplementary Materials

## Appendix A. Robustness of Aggregate-Based Distance-Decay Recovery

### A.1 Estimation Details

The distance decay parameters $(\gamma, \beta)$ are estimated via maximum likelihood estimation (MLE) from the aggregate trip-length distribution (marginal distance histogram) of source cities. To ensure numerical stability and enforce parameter bounds ($\beta > 0, \gamma \geq 0$), we perform L-BFGS-B optimization in log-space, parameterizing the variables as $\tilde{\beta} = \log(\beta)$ and $\tilde{\gamma} = \log(\gamma)$. The optimization employs a multi-restart framework with 10 random perturbations initialized around baseline data-driven anchors to prevent the solver from stalling in local optima, retaining the parameter set that maximizes log-likelihood.

### A.2 Robustness Analysis

We evaluate the robustness of our MLE parameter recovery under non-ideal data scenarios:

*Bin Sensitivity Analysis* Sweeping the distance bins $K \in \{10, 20, 50, 100\}$ shows that the recovery error for the power-law exponent $\gamma$ stabilizes at a low level (MAE $\approx 0.13$) for $K \geq 50$. This indicates that 50 percentile bins are sufficient to robustly capture the decay profile.
*Missing Attractiveness Data ($A_j$)* Randomly removing 10%, 20%, and 50% of target-city destination attractions $A_j$ yields highly stable decay recovery (e.g., $\gamma_{\text{MAE}} = 0.1312$ under 20% missing data vs. 0.1313 baseline). This validates the robustness of aggregate MLE against localized data gaps.
*CBD Collapse (Extreme $A_j$ Distortion)* Setting the attraction $A_j$ of the top 5% highest attractiveness zones to zero still allows accurate parameter recovery ($\gamma_{\text{MAE}} = 0.1225$). This proves that spatial decay friction can be recovered even when key destination hubs are omitted.

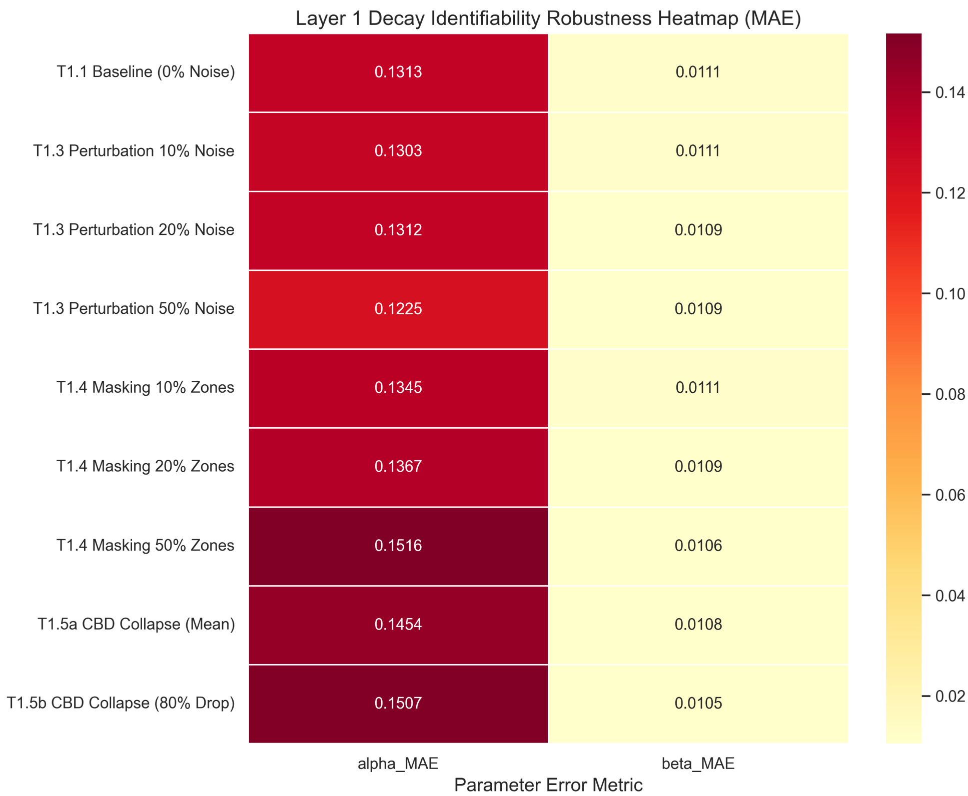


**Fig. 8** Robustness of Aggregate-Based Distance-Decay Recovery (MAE) under varying levels of noise, missing data, and CBD collapse scenarios across all 25 held-out target cities.

# Appendix B. Robustness of Origin Production Transfer

The origin trip production ($O_i$) estimator is trained using a Gradient Boosted Decision Tree (GBDT) regressor on source training cities. Complete hyperparameter configurations and implementation scripts are available in the project's code repository. Here, we analyze the sensitivity of the zero-shot production transfer.

## B.1 Source-Scale Sensitivity

We analyze how the number of training environments affects zero-shot predictability by retraining the GBDT on subsets of $n_{\text{src}} \in \{5, 10, 15, 20, 25\}$ source cities. Target-city evaluation shows that zero-shot prediction performance rises steeply from $n_{\text{src}} = 5$ and saturates around 15–20 source cities. This indicates that only a modest number of source cities is required to learn stable, generalizable outflow characteristics.

## B.2 Noise Sensitivity

To evaluate the stability of origin transfer under prediction errors, we inject additive Gaussian noise at log-scale ($\sigma \in \{10\%, 20\%, 40\%\}$) directly onto the predicted outflows $\log \hat{O}_i$. Even under substantial injected noise (40%), the target CPC degrades only marginally (from 0.7037 to 0.6923). This demonstrates that production-constraint normalization effectively dampens localized estimation noise.

# Appendix C. Feature Selection Sensitivity

## C.1 Feature Count Sensitivity

Evaluating the GBDT model performance across nested subsets of top SHAP features ($K \in \{5, 10, 15, 20, 22, 25, 30\}$) shows that predictive performance plateaus between 15 and 22 features (CPC: 0.7124). This indicates that most transferable spatial information is captured by a

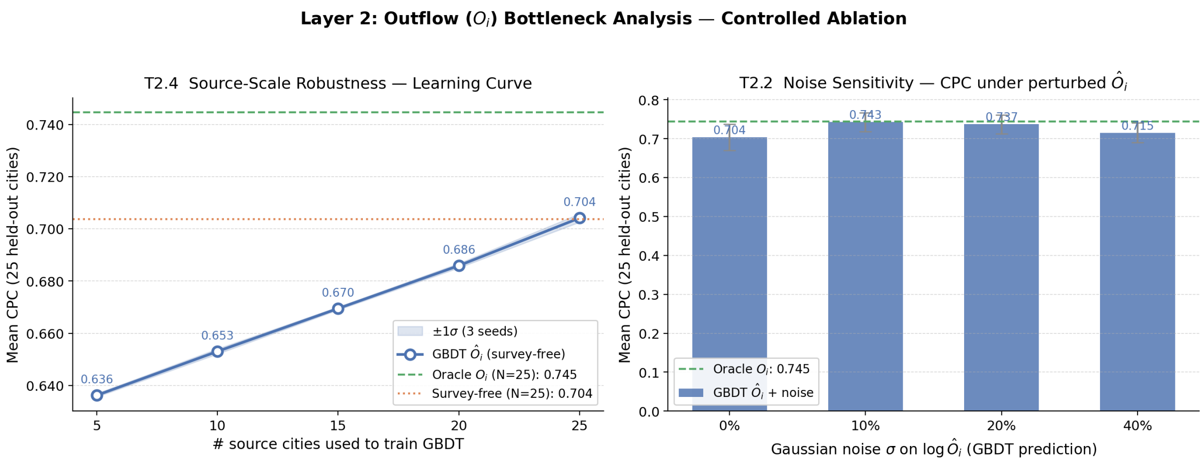


**Fig. 9** Outflow bottleneck analysis. **Left**: CPC vs. training source cities ($n_{\mathrm{src}}$), showing performance saturation at 15–20 cities. **Right**: Noise sensitivity of CPC under additive Gaussian noise $\sigma$ injected on predicted outflows $\log \hat{O}_i$.

compact set of built-environment indicators, with additional features providing marginal utility.

## A.6 List of Abbreviations

The key abbreviations and acronyms used throughout this paper are summarized and defined in Table 11.

# References

Alis C, Legara EFT, Monterola C (2021) Generalized radiation model for human migration. Scientific Reports 11(1):22707. https://doi.org/10.1038/s41598-021-02109-1

Atwal KS, Anderson T, Pfoser D, et al (2025) Commuting flow prediction using OpenStreetMap data. Computational Urban Science 5(1):2. https://doi.org/10.1007/s43762-025-00161-5

Barbosa H, Barthelemy M, Ghoshal G, et al (2018) Human mobility: Models and applications. Physics Reports 734:1–74. https://doi.org/10.1016/j.physrep.2018.01.001

Brockmann D, Hufnagel L, Geisel T (2006) The scaling laws of human travel. Nature 439(7075):462–465. https://doi.org/10.1038/nature04292

Gallotti R, Maniscalco D, Barthelemy M, et al (2024) Distorted insights from human mobility data. Communications Physics 7(1):1–10. https://doi.org/10.1038/s42005-024-01909-x

González MC, Hidalgo CA, Barabási AL (2008) Understanding individual human mobility patterns. Nature 453(7196):779–782. https://doi.org/10.1038/nature06958

Jin G, Wu Y, Wang M, et al (2022) Selective cross-city transfer learning for traffic prediction via source city region re-weighting. In: Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining. Association for Computing Machinery, pp 729–739, https://doi.org/10.1145/3534678.3539345

Lenormand M, Huet S, Gargiulo F, et al (2012) A universal model of commuting networks. PLoS One 7(10):e45985. https://doi.org/10.1371/journal.pone.0045985

Lenormand M, Bassolas A, Ramasco J (2016) Systematic comparison of trip distribution laws and models. Journal of Transport Geography 51:158–169. https://doi.org/10.1016/j.jtrangeo.2016.02.003

Lundberg SM, Lee SI (2017) A unified approach to interpreting model predictions. Advances in Neural Information Processing Systems 30:4765–4774. https://doi.org/10.48550/arXiv.1705.07874

**Table 10** GBDT Model Hyperparameter Configurations

| Hyperparameter | Outflow Model ($O_i$) | Attraction Model ($A_j$, ablation only) |
|---|---|---|
| Base Learner | Decision Tree | Decision Tree |
| Loss Function | Least Squares (MSE on $\log O_i$) | Least Squares (MSE on $A_j$) |
| Learning Rate ($\eta$) | 0.05 | 0.05 |
| Number of Estimators | 150 | 200 |
| Max Tree Depth | 6 | 6 |
| Num. Leaves | 31 | 31 |
| Subsample Ratio | 0.8 | 0.8 |
| Colsample By Tree | 0.8 | 0.8 |
| Minimum Child Weight | 1 | 1 |
| Early Stopping Rounds | 15 | 15 |

**Table 11** List of Abbreviations

| Abbreviation | Definition |
|---|---|
| CBD | Central Business District |
| CPC | Common Part of Commuters |
| GBDT | Gradient Boosted Decision Tree |
| GDPR | General Data Protection Regulation |
| LOCO | Leave-One-City-Out |
| MLE | Maximum Likelihood Estimation |
| MSE | Mean Squared Error |
| OD | Origin-Destination |
| OSM | OpenStreetMap |
| POI | Point of Interest |
| SDF-ZTF | Separable Decomposed Framework for Zero-Shot Transferable Flows |
| SHAP | SHapley Additive exPlanations |


Mai G, Cundy C, Choi K, et al (2022) Towards a foundation model for geospatial artificial intelligence. ACM Transactions on Intelligent Systems and Technology 13(6):1–25. https://doi.org/10.1145/3557915.3561043

Mendieta M, Han B, Shi X, et al (2023) Towards geospatial foundation models via continual pretraining. In: Proceedings of the IEEE/CVF International Conference on Computer Vision, pp 16781–16790, https://doi.org/10.1109/ICCV53024.2023.01529

Meta AI (2026) Movement distribution maps. https://ai.meta.com/ai-for-good/datasets/movement-distribution-maps, accessed: 2026-06-20

Pappalardo L, Manley E, Sekara V, et al (2023) Future directions in human mobility science. Nature Computational Science 3:588–600. https://doi.org/10.1038/s43588-023-00469-4

Rong C, Feng J, Ding J, et al (2023) GODDAG: Generating origin-destination flow for new cities via domain adversarial training. IEEE Transactions on Knowledge and Data Engineering 35(10):10048–10057. https://doi.org/10.1109/TKDE.2022.3218526

Simini F, Gonzalez M, Maritan A, et al (2012) A universal model for mobility and migration patterns. Nature 484:96–100. https://doi.org/10.1038/nature10856

Simini F, Barlacchi G, Luca M, et al (2021) A deep gravity model for mobility flows generation. Nature Communications 12(1):5647. https://doi.org/10.1038/s41467-021-26752-6

Song C, Qu Z, Blumm N, et al (2010) Limits of predictability in human mobility. Science 327(5968):1018–1021. https://doi.org/10.1126/science.1177170

Stouffer SA (1940) Intervening opportunities: A theory relating mobility and distance. American Sociological Review 5(6):845–867. https://doi.org/10.2307/2084520

Tanner J (1961) Factors affecting the amount of travel. Road Research Technical Paper 51, Road Research Laboratory, London

Tatem AJ (2017) WorldPop, open data for spatial demography. Scientific Data 4:170004. https://doi.org/10.1038/sdata.2017.4, worldPop gridded population datasets available at https://www.worldpop.org/

Wang L, Geng X, Ma X, et al (2018) Crowd flow prediction by deep spatio-temporal transfer learning. In: Proceedings of the 27th International Joint Conference on Artificial Intelligence, pp 1341–1347, https://doi.org/10.24963/ijcai.2018/186

Wang S, Cao J, Yu PS (2021) Deep learning based origin-destination prediction via contextual information fusion. Multimedia Tools and Applications 80(20):30427–30449. https://doi.org/10.1007/s11042-020-09355-6

Wang X, Jin K, Sun L, et al (2022) When transfer learning meets cross-city urban flow prediction: Spatio-temporal adaptation matters. In: Proceedings of the Thirty-First International Joint Conference on Artificial Intelligence (IJCAI-22), pp 2219–2225, https://doi.org/10.24963/ijcai.2022/308

Wilson A (1971) A family of spatial interaction models, and associated developments. Environment and Planning A 3(1):1–32. https://doi.org/10.1068/a030001

Yao H, Liu Y, Wei Y, et al (2019) Learning from multiple cities: A meta-learning approach for spatial-temporal prediction. In: Proceedings of the Web Conference 2019, pp 2123–2129, https://doi.org/10.1145/3308558.3313577